

## 2. Algebra

### *Notation, vocabulary and manipulation*

What students need to learn:

- A1** use and interpret algebraic manipulation, including:
- $ab$  in place of  $a \times b$
  - $3y$  in place of  $y + y + y$  and  $3 \times y$
  - $a^2$  in place of  $a \times a$ ,  $a^3$  in place of  $a \times a \times a$ ,  $a^2b$  in place of  $a \times a \times b$
  - $\frac{a}{b}$  in place of  $a \div b$
  - coefficients written as fractions rather than as decimals
  - brackets
- A2** substitute numerical values into formulae and expressions, including scientific formulae
- A3** understand and use the concepts and vocabulary of expressions, equations, formulae, identities, inequalities, terms and factors

- A4** simplify and manipulate algebraic expressions (including those involving surds and algebraic fractions) by:
- collecting like terms
  - multiplying a single term over a bracket
  - taking out common factors
  - expanding products of two or more binomials
  - factorising quadratic expressions of the form  $x^2 + bx + c$ , including the difference of two squares; **factorising quadratic expressions of the form  $ax^2 + bx + c$**
  - simplifying expressions involving sums, products and powers, including the laws of indices
- A5** understand and use standard mathematical formulae; rearrange formulae to change the subject
- A6** know the difference between an equation and an identity; argue mathematically to show algebraic expressions are equivalent, and use algebra to support and construct arguments and proofs
- A7** where appropriate, interpret simple expressions as functions with inputs and outputs; **interpret the reverse process as the 'inverse function'; interpret the succession of two functions as a 'composite function' (the use of formal function notation is expected)**

### Graphs

What students need to learn:

- A8** work with coordinates in all four quadrants
- A9** plot graphs of equations that correspond to straight-line graphs in the coordinate plane; use the form  $y = mx + c$  to identify parallel and perpendicular lines; find the equation of the line through two given points or through one point with a given gradient
- A10** identify and interpret gradients and intercepts of linear functions graphically and algebraically
- A11** identify and interpret roots, intercepts, turning points of quadratic functions graphically; deduce roots algebraically **and turning points by completing the square**
- A12** recognise, sketch and interpret graphs of linear functions, quadratic functions, simple cubic functions, the reciprocal function  $y = \frac{1}{x}$  with  $x \neq 0$ ,  
**exponential functions  $y = k^x$  for positive values of  $k$ , and the trigonometric functions (with arguments in degrees)  $y = \sin x$ ,  $y = \cos x$  and  $y = \tan x$  for angles of any size**
- A13** **sketch translations and reflections of a given function**

- A14** plot and interpret graphs (including reciprocal graphs and exponential graphs) and graphs of non-standard functions in real contexts to find approximate solutions to problems such as simple kinematic problems involving distance, speed and acceleration
- A15** **calculate or estimate gradients of graphs and areas under graphs (including quadratic and other non-linear graphs), and interpret results in cases such as distance-time graphs, velocity-time graphs and graphs in financial contexts (this does not include calculus)**
- A16** **recognise and use the equation of a circle with centre at the origin; find the equation of a tangent to a circle at a given point**

### *Solving equations and inequalities*

What students need to learn:

- A17** solve linear equations in one unknown algebraically (including those with the unknown on both sides of the equation); find approximate solutions using a graph
- A18** solve quadratic equations (including those that require rearrangement) algebraically by factorising, by completing the square and by using the quadratic formula; find approximate solutions using a graph
- A19** solve two simultaneous equations in two variables (linear/linear or linear/quadratic) algebraically; find approximate solutions using a graph
- A20** **find approximate solutions to equations numerically using iteration**
- A21** translate simple situations or procedures into algebraic expressions or formulae; derive an equation (or two simultaneous equations), solve the equation(s) and interpret the solution
- A22** solve linear inequalities in one or two variable(s), and quadratic inequalities in one variable; represent the solution set on a number line, using set notation and on a graph

### *Sequences*

What students need to learn:

- A23** generate terms of a sequence from either a term-to-term or a position-to-term rule
- A24** recognise and use sequences of triangular, square and cube numbers, simple arithmetic progressions, Fibonacci type sequences, quadratic sequences, and simple geometric progressions ( $r^n$  where  $n$  is an integer, and  $r$  is a rational number  $> 0$  or a surd) **and other sequences**
- A25** deduce expressions to calculate the  $n$ th term of linear **and quadratic** sequences